General description

The pressure regulation arrangement, type 2", is designed for incorporation in LPG piping installations as close to the place of filling as possible and is used to maintain the desired constant filling pressure in an LPG filling system. A constant filling pressure is a prerequisite to achieve accurate filling tolerances on the filling system.

The arrangement is mainly consisting of a filter to protect the gas stop valves and filling heads on the filling machines in the filling system, a differential pressure valve and gas stop valves used during maintenance and servicing of the differential pressure valve.

The differential pressure valve works as a pressure regulation valve (an overflow valve) and maintains the desired filling pressure independently of the number of active filling machines in the filling system.

The pressure regulation process is carried out as follows: the differential pressure valve leads a large or small quantity of gas back to the storage tank via a return line thereby maintaining a constant pressure in the supply line to the filling system. The pre-set pressure can be regulated by means of a spring-loaded piston in the differential pressure valve.

IMPORTANT! It is recommnded to place the pressure regulation arrangement as close to the place of filling as possible.

The pressure regulation arrangement type 2" consists of the following main components:

- 2" differential pressure valve with counter flanges, gaskets and bolts according to ANSI standard
- 3 gas stop valves, 1 exhaust valve and 1 safety valve
- Filter
- Manometer and manometer cock
- Necessary pipe fittings, flanges, gaskets, bolts, equalizing connections (copper strips), etc.

Scope of supply

The pressure regulation arrangement is supplied unassembled for incorporation in an LPG piping installation with corresponding 2" differential pressure valve with counter flanges, gaskets and bolts according to ANSI standard, 3 gas stop valves, 1 exhaust valve, 1 safety valve, filter, manometer, manometer cock and necessary pipe fittings, flanges, gaskets, bolts, equalizing connections (copper strips), etc.

Additional equipment

(Only included if specified in quotation)

- Standard spare parts set for differential pressure valve

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Technical data

Capacity and conditions	• From pump to filling system via pressure regulation arrangement: Max. 20 m³/hour • From pump back to storage tank via differential pressure valve: Max. 600 litres/minute
Necessary connections (specifications and quality requirements)	• LPG piping installation: – 2" supply line from pump – 2" return line to storage tank – 2" supply line to filling system • Earth net: – When assembling the pressure regulation arrangement the supplied equalizing connections, which have a transition resistance of ≤ 2.0 Ohm, must be mounted at the flanged joints. The pressure regulation arrangement and the corresponding piping installation must be earthed as required according to EN 60 204-1 or API RP 2003
Weight	• Approx. 250 kg
Other data	• Regulation range: From 0 to 1.2 Mpa. (0-12 bar) • Function tolerance: ± 0.1 Mpa. (± 1 bar) • Recommended differential pressure: From 0.7 to 0.9 Mpa. (7-9 bar) • Working pressure: Max. 2.1 Mpa. (21 bar) • Test pressure (without safety valve): Max. 3.0 Mpa. (30 bar) • The pressure regulation arrangement is designed for installation in places in which the temperature may vary between -10°C and +40°C

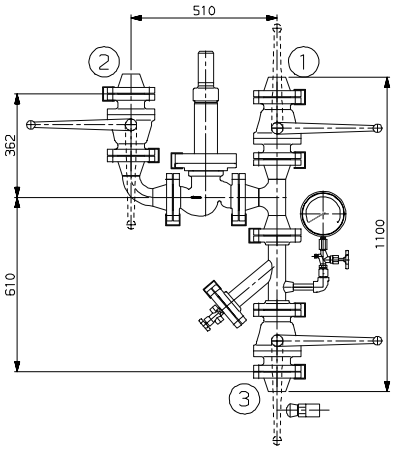
Material specifications

Description	Designation	Material	Execution
Steel pipes	ANSI B36.10 SCH 40	API Std 5L-GR.B	Seamless
Flanges	ANSI B16.5	ASTM-A105 GR.2	WN-RF-300 lbs
Welding sockets	ANSI B16.11	ASTM-A105 GR.2	Thredolets
Bolts	ANSI B16.5	ASTM-193-GR.B7	Stud bolts
Nuts	ANSI B16.5	ASTM-A194-GR.2H	
Gaskets	ANSI B16.21	Kenovit S	Flat ring

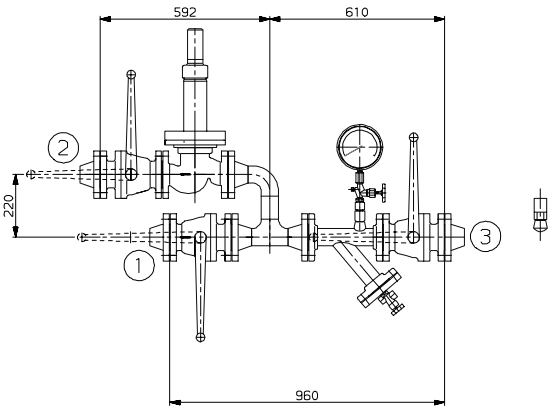
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Dimensions in mm

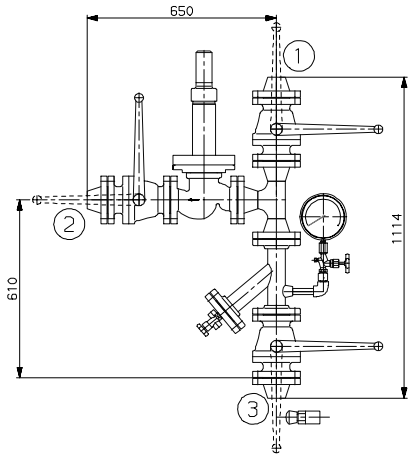
Type A



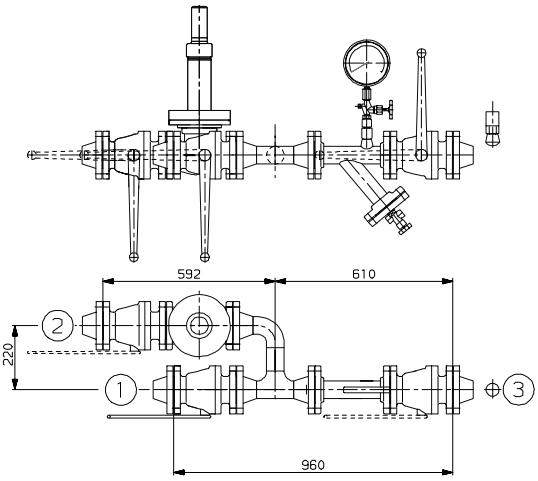
Type B



Type C



Type D



1 Pipe connection to 2" supply line from pump

2 Pipe connection to 2" return line to storage tank

3 Pipe connection to 2" supply line to filling system

The pressure regulation arrangement is supplied unassembled and the components can be assembled in different ways (e.g. types A, B, C and D) so that the pressure regulation arrangement fit the actual piping installation.

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